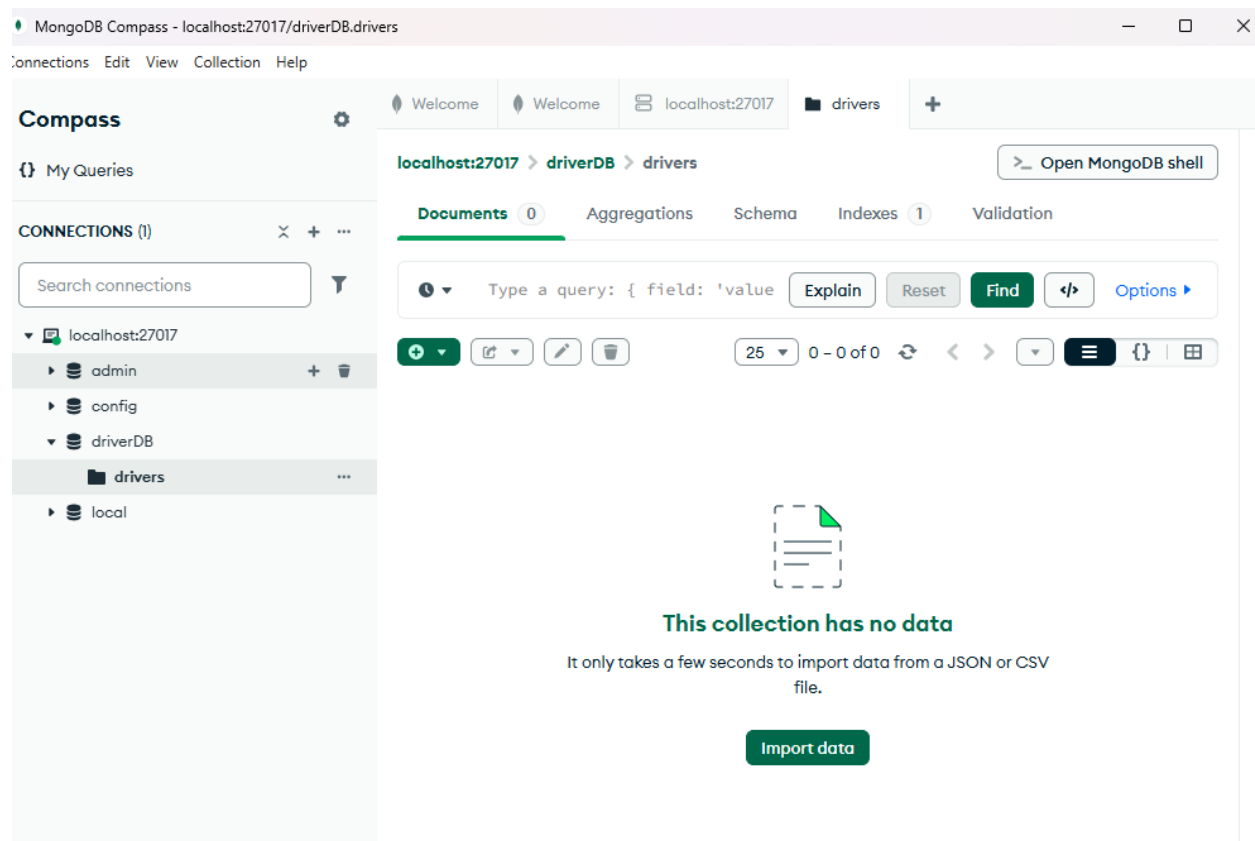
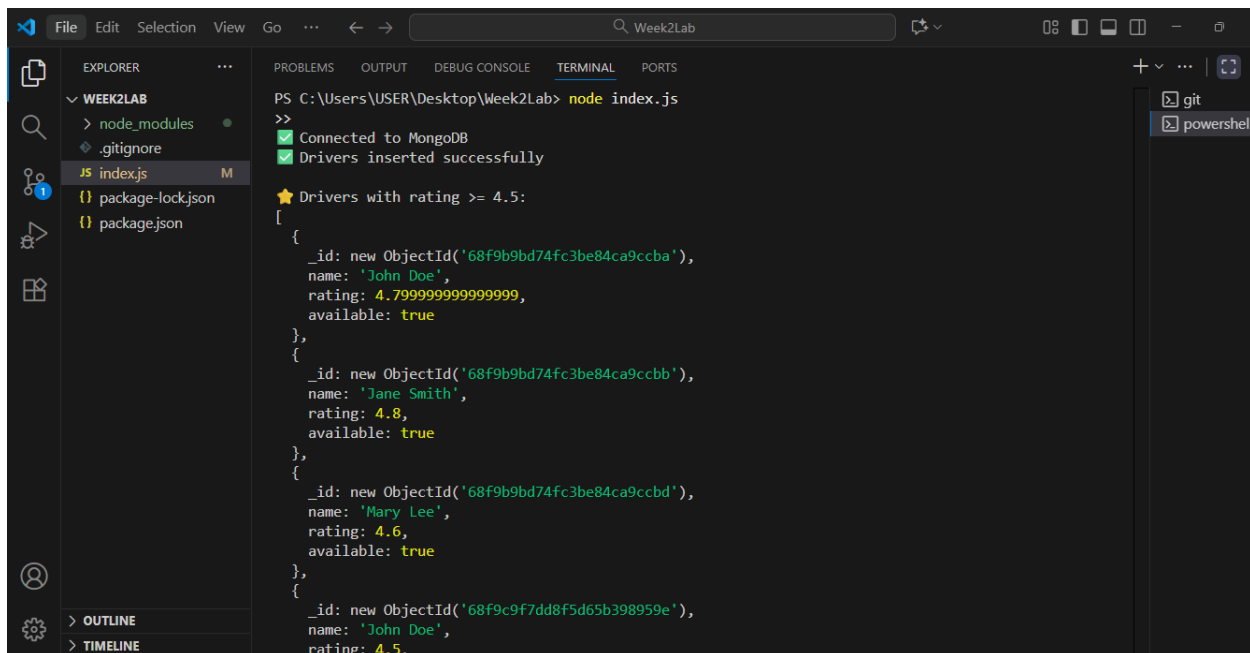
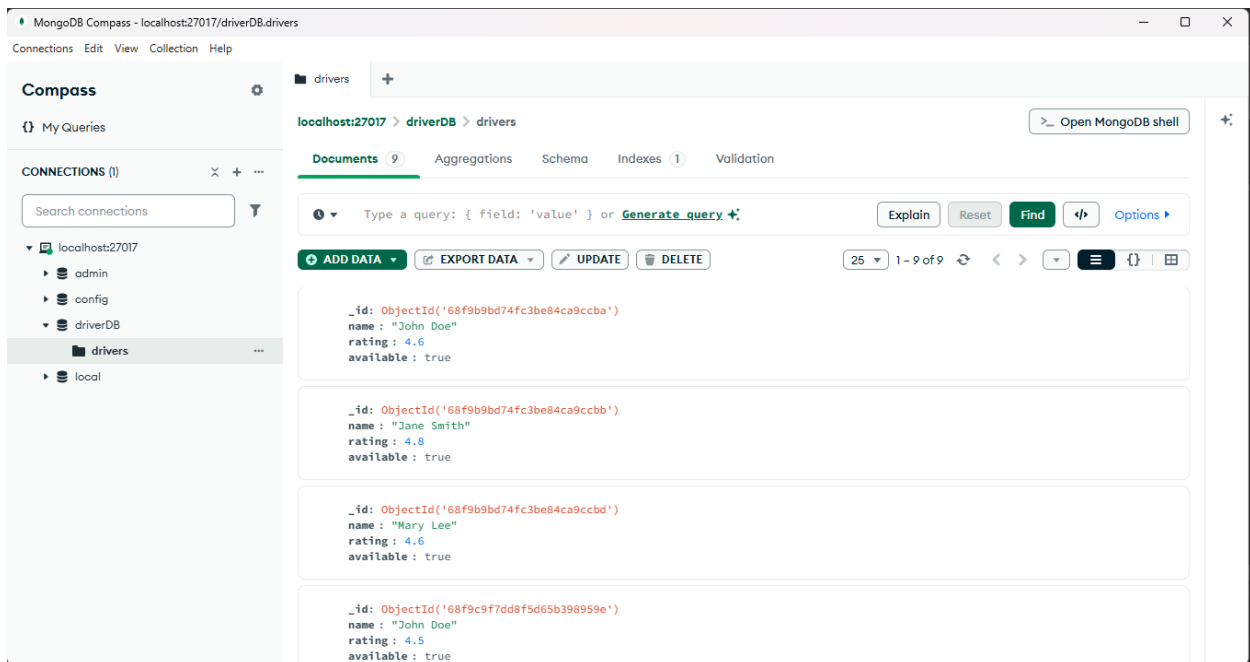


NAME:RIGNES TAMIL VANAN

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WEEK 2 LAB





```
rating: 4.6,
available: true
},
{
  _id: new ObjectId('68f9c9f7dd8f5d65b398959e'),
  name: 'John Doe',
  rating: 4.5,
  available: true
},
{
  _id: new ObjectId('68f9c9f7dd8f5d65b398959f'),
  name: 'Jane Smith',
  rating: 4.8,
  available: true
},
{
  _id: new ObjectId('68f9c9f7dd8f5d65b39895a1'),
  name: 'Mary Lee',
  rating: 4.6,
  available: true
},
{
  _id: new ObjectId('68f9ce664acff10057d569d3'),
  name: 'John Doe',
  rating: 4.5,
  available: true
},
{
  _id: new ObjectId('68f9ce664acff10057d569d4'),
```

```
available: true
}
]

Increased John Doe's rating by 0.1

Deleted unavailable drivers

Connection closed
```

```
_id: new ObjectId('68f9c9f7dd8f5d65b39895a1'),
name: 'Mary Lee',
rating: 4.6,
available: true
}
]

Increased John Doe's rating by 0.1

Deleted unavailable drivers

Connection closed
PS C:\Users\USER\Desktop\Week2Lab>
```

1.Explain CRUD operations and relation to MongoDB functions

- CRUD = Create, Read, Update, Delete
- Create → insertMany() → adds sample drivers to the collection
- Read → find(), findOne() → fetches drivers with rating ≥ 4.5; checks John Doe's updated rating
- Update → updateOne() → increases John Doe's rating by 0.1

- Delete → deleteMany() → removes drivers with available: false
- Summary: CRUD operations let you manage database data (add, view, modify, remove)

2. MongoDB operators used and their usage

- \$gte → { rating: { \$gte: 4.5 } } → finds documents with rating $\geq$ 4.5
- \$inc → { \$inc: { rating: 0.1 } } → increases numeric field (rating) by 0.1
- \$set → { \$set: { rating: 4.5 } } → sets a field to a specific value (used to reset rating)
- Filter { available: false } → used in deleteMany() to match unavailable drivers

3. Before: updateOne({ name: "John Doe" }, { \$inc: { rating: 0.1 } })

- Only the first John Doe document would increase by 0.1.

```
// Ensure John Doe starts at 4.5, then increase to 4.6
await drivers.updateOne(
  { name: "John Doe" },
  { $set: { rating: 4.5 } }
);
await drivers.updateOne(
  { name: "John Doe" },
  { $inc: { rating: 0.1 } }
);
```

After: updateMany({ rating: { \$gte: 4.5 } }, { \$inc: { rating: 0.1 } })

```
40      await drivers.updateOne(
41        { name: "John Doe" },
42        { $set: { rating: 4.5 } }
43      );
44      await drivers.updateMany(
45        { rating: { $gte: 4.5 } },
46        { $inc: { rating: 0.1 } }
47      );
48
```

- All documents with rating $\geq$ 4.5 would increase.
- Console shows all top-rated drivers updated.
- Compass shows multiple rows updated, not just John Doe.
- Multiple John Doe / Jane Smith / Mary Lee documents exist.
- updateMany would increase all matches if rerun.
- Currently, after your last run, John Doe stays at 4.6 because the update line sets it consistently.
- updateOne → updates only first match (John Doe)
- updateMany → updates all matching drivers (rating $\geq$ 4.5)
- Console shows multiple updates for updateMany
- Compass shows multiple rows changed

- Conclusion: updateMany affects all matches, updateOne only the first match

4. Before: deleteOne({ available: false }) → deletes only first unavailable driver

After: deleteMany({ available: false }) → deletes all unavailable drivers

```
}  
  
Removed 0 unavailable driver(s) 🗑️  
  
MongoDB connection closed 🔒  
PS C:\Users\USER> cd .\MongoDB\ & node index.js
```